

Examination : **End Semester Examination Nov/Dec 2025**

Name of the Course : **B.Tech. (Information Technology & Mathematical Innovation)**

Unique Paper Code : **3122613501**

Name of the Paper : **Linear Programming and Game Theory (DSC13)**

Semester : **V**

Duration : **3 hours**

Maximum Marks : **90**

Instruction: Question No. 1 is compulsory.

1. A factory produces two types of molded plastic components, P1 and P2, using three machines: M1, M2, and M3. Each product requires machine time on all three machines, and each machine has a fixed number of available hours per week. The data on machine time requirements and profit per unit are given below

Machine	Hours required per unit		Available Hours
	P1	P2	
M1	2	4	100
M2	3	2	90
M3	1	3	72
Profit/unit	60	80	

After solving this LP problem, its optimal simplex tableau is

Basic	x_1	x_2	s_1	s_2	s_3	b
x_1	1	0	1/2	0	0	18
x_2	0	1	-3/8	1/4	0	16
z	0	0	7.5	10	0	2160
s_3	0	0	-5/8	2/3	1	16/3

Suppose the market for the product 1 i.e. P1, changes and its profit increases from Rs. 60 to Rs. 75 per unit. Determine whether the current production plan remains optimal and if not find the new optimal solution. [12]

Section 1

Attempt any 3 questions from this section. Each question carries 12 marks. [3x12=36]

- (I). The city of Erstville is faced with a severe budget shortage. Seeking a long-term solution, the city council votes to improve the tax base by condemning an inner-city housing area and

replacing it with a modern development. The project involves two phases: (1) demolishing substandard houses to provide land for the new development and (2) building the new development. The following is a summary of the situation.

1. As many as 300 substandard houses can be demolished. Each house occupies a .25-acre lot. The cost of demolishing a condemned house is \$2000.
2. Lot sizes for new single-, double-, triple-, and quadruple-family homes (units) are .18, .28, .4, and .5 acre, respectively. Streets, open space, and utility easements account for 15% of available acreage.
3. In the new development, the triple and quadruple units account for at least 25% of the total. Single units must be at least 20% of all units, and double units at least 10%.
4. The tax levied per unit for single, double, triple, and quadruple units is \$1000, \$1900, \$2700, and \$3400, respectively.
5. The construction cost per unit for single-, double-, triple-, and quadruple-family homes is \$50,000, \$70,000, \$130,000, and \$160,000, respectively.
6. Financing through a local bank is limited to \$15 million.

Develop a linear model to maximize tax collection for the city of Erstville.

(II). Periodic preventive maintenance is carried out on aircraft engines, where an important component must be replaced. The numbers of aircraft scheduled for such maintenance over the next six months are estimated at 200, 180, 300, 198, 230, and 290, respectively. All maintenance work is done during the first day of the month, where a used component may be replaced with a new or an overhauled component. The overhauling of used components may be done in a local repair facility, where they will be ready for use at the beginning of next month, or they may be sent to a central repair shop, where a delay of 3 months (including the month in which maintenance occurs) is expected. The repair cost in the local shop is \$120 per component. At the central facility, the cost is only \$35 per component. An over-hauled component used in a later month will incur an additional storage cost of \$1.50 per unit per month. New components may be purchased at \$200 each in month 1, with a 5% price increase every 2 months. Formulate the problem as a transportation model.

- (III). Two companies promote two competing products. Currently, each product controls 50% of the market. Because of recent improvements in the two products, each company plans to launch an advertising campaign. If neither company advertises, equal market shares will continue. If either company launches a stronger campaign, the other company is certain to lose a proportional percentage of its customers. A survey of the market shows that 50% of potential customers can be reached through television, 30% through newspapers, and 20% through radio.

- (a) Formulate the problem as a two-person zero-sum game, and determine the advertising media for each company. [7]
- (b) Determine a range for the value of the game. [5]

(IV). Fox Enterprises is considering six projects for possible construction over the next four years. Fox can undertake any of the projects partially or completely. A partial undertaking of a

project will prorate both the return and cash outlays proportionately. The expected (present value) returns and cash outlays for the projects are given in the following table.

Project	Cash outlay (\$1000)				Return (\$1000)
	Year 1	Year 2	Year 3	Year 4	
1	10.5	14.4	2.2	2.4	324.00
2	8.3	12.6	9.5	3.1	358.00
3	10.2	14.2	5.6	4.2	177.50
4	7.2	10.5	7.5	5.0	148.00
5	12.3	10.1	8.3	6.3	182.00
6	9.2	7.8	6.9	5.1	123.50
Available funds (\$1000)	60.0	70.0	35.0	20.0	

Formulate the problem as a linear program. Suppose that if a portion of project 2 is undertaken, then at least an equal portion of project 6 must be undertaken. Modify the formulation of the model.

Section 2

Attempt any 3 questions from this section. Each question carries 16 marks. [3x14=42]

- (I). JoShop wants to assign four different categories of machines to five types of tasks. The numbers of machines available in the four categories are 25, 30, 20, and 30. The numbers of jobs in the five tasks are 30, 10, 20, 25, and 20. Machine category 4 cannot be assigned to task type 4. Given table provides the unit cost (in dollars) of assigning a machine category to a task type. The objective of the problem is to determine the optimum number of machines in each category to be assigned to each task type. Solve the problem and interpret the solution.

		Task type				
		1	2	3	4	5
Machine category	1	10	2	3	15	9
	2	5	10	15	2	4
	3	15	5	14	7	15
	4	20	15	13	—	8

- (II). Three refineries with daily capacities of 6, 5, and 8 million gallons, respectively, supply three distribution areas with daily demands of 4, 8, and 7 million gallons, respectively. Gasoline is transported to the three distribution areas through a network of pipelines. The transportation cost is 10 cents per 1000 gallons per pipeline mile. Table (given below) gives the mileage between the refineries and the distribution areas. Refinery 1 is not connected to distribution 3.

- (a) Construct the associated transportation model. [6]
 (b) Determine the optimum shipping schedule in the network (perform 2 iterations of the method). [8]

	Distribution area		
	1	2	3
Refinery 1	180	180	—
Refinery 2	300	800	900
Refinery 3	220	200	120

(III). The Burroughs Garment Company manufactures men's shirts and women's blouses for Walmark Discount Stores. Walmark will accept all the production supplied by Burroughs. The production process includes cutting, sewing, and packaging. Burroughs employs 25 workers in the cutting department, 35 in the sewing department, and 5 in the packaging department. The factory works one 8-hr shift, 5 days a week. The following table gives the time requirements and prices per unit for the two garments:

Garment	Minutes per unit			Unit price (\$)
	Cutting	Sewing	Packaging	
Shirts	20	70	12	8.00
Blouses	60	60	4	12.00

- (a) Determine the optimal weekly production schedule for Burroughs. [6]
 (b) Determine the worth of 1 hr of cutting, sewing, and packaging in terms of the total revenue. [4]
 (c) If overtime can be used in cutting and sewing, what is the maximum hourly rate Burroughs should pay for overtime? [4]

(IV). Wild West produces two types of cowboy hats. A Type 1 hat requires twice as much labor time as a Type 2. If all the available labor time is dedicated to Type 2 alone, the company can produce a total of 400 Type 2 hats a day. The respective market limits for the two types are 150 and 200 hats per day. The revenue is \$8 per Type 1 hat and \$5 per Type 2 hat.

- (a) Determine the number of hats of each type that maximizes revenue. [6]
 (b) Determine the dual price of the production capacity (in terms of the Type 2 hat) and the range for which it is applicable. [4]
 (c) If the daily demand limit on the Type 1 hat is decreased to 120, use the shadow price to determine the corresponding effect on the optimal revenue. [4]